

Gauss–Hurwitz Pole Cancellation

Resonant gamma-ratio sums, harmonic identities,
Stieltjes jets, and polylogarithmic endpoint subtraction

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Abstract

We construct a common exact calculus for a Gauss gamma-ratio series and its Hurwitz-zeta counterterms. Subtracting the first K explicit asymptotic coefficients gives an ordinary normally convergent series on $\operatorname{Re} u > -K$. At every nonpositive integer excess $u = -N$, its value is

$$\frac{(-1)^N (1-a)_N (1-b)_N}{N!} (H_N - \gamma - \psi(a) - \psi(b) + \psi(c)).$$

A coefficient differential identity explains the otherwise substantial Bernoulli-polynomial cancellation. An all-order Taylor formula then expresses every resonant spectral derivative in a finite collection of generalized Stieltjes constants, nonpositive-integer Hurwitz-zeta derivatives, and gamma/polygamma data. Reciprocal-gamma zeros are handled by entire Taylor expansions; terms vanishing at resonance can contribute to the first and subsequent derivatives.

A second consequence is an all-weight identity for gamma-weighted complete harmonic tails: the entire family collapses to $2\zeta(2k+1, x)$. In the central-binomial-square case this gives, for example, a trigamma-tail sum of value $14\zeta(3)$ and a quadratic-tail sum of value $62\zeta(5)$. We also prove an exact telescoping antiderivative formula and a polylogarithmic endpoint-subtraction theorem, including all spectral jets.

The underlying Gauss summation and hypergeometric constant-term theory are classical. Their overlap with Bühring’s work is explicitly acknowledged; no global priority claim is made. The contribution here is a self-contained all-resonance, all-jet synthesis with exact coefficient recurrences, explicit identities, and reproducible checks for integration into ProveIt.

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1 Scope, provenance, and conventions

1.1 What this continuation establishes

The requested direction is exact special-function identities, rather than inequalities or numerical nonreduction. The present development connects five objects: gamma ratios, generalized harmonic numbers, spectral Hurwitz-zeta derivatives, generalized Stieltjes constants, and polylogarithmic endpoint counterterms. Asymptotic expansions enter only to prove that certain explicitly displayed differences are genuinely summable and that differentiation is legitimate. The final statements are identities.

There are three main mechanisms. First, the poles of Gauss’s summation formula cancel the poles of the matching Hurwitz-zeta counterterms. Second, the asymptotic coefficients satisfy a parameter differential law, which collapses every zeroth-order resonance to a short digamma expression. Third, Taylor coefficients preserve the cancellation and produce Stieltjes constants at precisely the colliding pole. A different, balanced parameter variation generates the all-weight harmonic-tail identity of Theorem 5.3.

The initial canonical repository README reports an exact proof of S_4 and keeps S_6 and the revised S_8 conjectural. It also distinguishes analytic proofs, finite exact certificates, and small numerical residuals [8]. Those distinctions are retained here. We do not reprove S_4 , promote either remaining Gaussian conjecture, or infer any independence statement from numerical searches.

The three inspected incoming packages on shifted Hurwitz jets and Stieltjes correlations already develop circular convolution, contact terms, and primitive ladders. Our construction instead starts from a hypergeometric coefficient sequence. The full incoming directory was changing during the investigation; the audit is not exhaustive. The latest observed repository head was `e0d9463bdee9685dfb1dddb819059cc738540c57`. Individual inspected files and limitations are recorded in `notes/SOURCE_AUDIT.md`. A small wording correction was rechecked at that exact commit; no repository file was modified.

1.2 Classical inputs and the priority boundary

Gauss’s ${}_2F_1(1)$ evaluation, Stirling–Bernoulli gamma-ratio expansions, and the defining identities for Hurwitz zeta are classical [1, 2, 3]. Bühring’s treatment of hypergeometric partial sums includes zero and negative integer excess and their constant terms [5]. In particular, the zeroth-order evaluation below belongs to that classical constant-term setting; its short coefficient proof is an independent organization, not a claim to have discovered hypergeometric finite parts. Our coefficients A_j are *not* Bühring’s coefficients $A_k^{(p)}$: the latter encode a different continuation expansion. The present fixed-shift subtraction basis, all-order Stieltjes coefficient formula, degeneracy-safe implementation, and combined primitive calculus are the specific development supplied to the repository.

Parameter differentiation of classical summations is likewise an established source of harmonic identities. We give complete proofs of our particular all-weight formulas, but do not claim that every specialization is absent from the literature. A full priority audit of individual binomial sums would require comparison with a much larger hypergeometric literature. No famous open problem is declared solved.

1.3 Notation and analytic domains

Unless a local complex extension is explicitly invoked, a, b, c, x are positive real numbers, N, m are nonnegative integers, and K is a positive integer. We use $H_0 = 0$ and

$$H_n^{(r)} = \sum_{j=1}^n \frac{1}{j^r}, \quad H_n = H_n^{(1)}, \quad (y)_N = \prod_{j=0}^{N-1} (y + j).$$

Empty products equal one. The falling factorial is $n^{\underline{N}} = n(n-1)\cdots(n-N+1)$. The generalized Stieltjes constants have the convention

$$\zeta(1+t, c) = \frac{1}{t} + \sum_{h=0}^{\infty} \frac{(-1)^h \gamma_h(c)}{h!} t^h, \quad \gamma_0(c) = -\psi(c). \quad (1.1)$$

We reserve $\zeta^{[q]}(s, c) = \partial_s^q \zeta(s, c)$ for derivatives in *order*; $\psi^{(q)}$ means a derivative in *argument*. A jet means an ordinary Taylor coefficient: $[t^m]f(t) = f^{(m)}(0)/m!$. This normalization matters in every all-order formula.

Our complete exponential Bell polynomials are specified by

$$\exp\left(\sum_{k \geq 1} X_k \frac{t^k}{k!}\right) = \sum_{m \geq 0} \mathcal{B}_m(X_1, \dots, X_m) \frac{t^m}{m!}. \quad (1.2)$$

Thus $\mathcal{B}_1 = X_1$, $\mathcal{B}_2 = X_1^2 + X_2$, and $\mathcal{B}_3 = X_1^3 + 3X_1X_2 + X_3$. Reciprocal gamma is always interpreted as the entire function $1/\Gamma$, not as a separately evaluated division by an infinite number.

2 The normally convergent Gauss–Hurwitz difference

2.1 The summand and its exact summation

Define

$$r_n(u) = \frac{\Gamma(n+a)\Gamma(n+b)}{\Gamma(n+1)\Gamma(n+a+b+u)}, \quad G(u) = \frac{\Gamma(a)\Gamma(b)\Gamma(u)}{\Gamma(a+u)\Gamma(b+u)}. \quad (2.1)$$

For $\operatorname{Re} u > 0$ one has

$$\sum_{n=0}^{\infty} r_n(u) = G(u). \quad (2.2)$$

Here is a direct proof, which also fixes the normalization. The beta integral gives

$$r_n(u) = \frac{\Gamma(n+a)}{\Gamma(n+1)\Gamma(a+u)} \int_0^1 t^{n+b-1} (1-t)^{a+u-1} dt.$$

Summing $\Gamma(n+a)t^n/n! = \Gamma(a)(1-t)^{-a}$ gives $\Gamma(a)B(b, u)/\Gamma(a+u) = G(u)$. For real positive parameters Tonelli's theorem applies; on compact subsets of $\operatorname{Re} u > 0$ the same calculation follows by absolute convergence and analytic continuation. This is Gauss's classical summation [1, 15.4.20].

2.2 Explicit coefficient polynomials

Let $B_k(y)$ denote the Bernoulli polynomial, with $B_1(y) = y - \frac{1}{2}$. For $k \geq 1$ put

$$\ell_k(u; c) = \frac{(-1)^{k+1}}{k(k+1)} \{B_{k+1}(a-c) + B_{k+1}(b-c) - B_{k+1}(1-c) - B_{k+1}(a+b+u-c)\}. \quad (2.3)$$

Define $A_0 = 1$ and, successively,

$$jA_j(u; c) = \sum_{k=1}^j k\ell_k(u; c)A_{j-k}(u; c). \quad (2.4)$$

These are polynomials with rational coefficients in a, b, u, c . The first nontrivial one is

$$A_1(u; c) = c - ab + (c - a - b + \tfrac{1}{2})u - \tfrac{1}{2}u^2. \quad (2.5)$$

The recurrence is a finite exact formula at every prescribed order; no infinite asymptotic expansion is numerically summed.

Lemma 2.1 (Uniform finite-order expansion). *For any fixed $K \geq 1$,*

$$r_n(u) = (n+c)^{-1-u} \left\{ \sum_{j=0}^{K-1} \frac{A_j(u; c)}{(n+c)^j} + O((n+c)^{-K}) \right\}. \quad (2.6)$$

The error is uniform on compact subsets of the stated parameter domain, including compact sets of complex u . Arbitrary fixed-order parameter and spectral derivatives have the corresponding uniform expansions; spectral differentiation introduces at most fixed powers of $\log(n+c)$.

Proof. Apply the finite Stirling expansion for $\log \Gamma(X+d)$, with $X = n+c$, to the two numerator and two denominator factors. The common $X \log X - X$ and constant terms cancel. The coefficient of $\log X$ is $-1-u$, and the inverse-power coefficients are exactly (2.3). Exponentiating a finite expansion with its remainder gives (2.6). Differentiating the formal exponential yields (2.4). Uniformity follows from the uniform remainder on compact sets of shifts in the Stirling expansion [2, §5.11]. For a fully analytic justification of derivatives, enlarge a given compact parameter set slightly, apply the uniform expansion there, and use Cauchy’s integral formula for the remainder. Only the explicit factor X^{-u} contributes the possible logarithmic powers. Finitely many small n are handled separately by entire reciprocal gamma. $\square$

Theorem 2.2 (Gauss–Hurwitz subtraction). *For $K \geq 1$, the series*

$$R_K(u; c) = \sum_{n=0}^{\infty} \left\{ r_n(u) - \sum_{j=0}^{K-1} \frac{A_j(u; c)}{(n+c)^{1+u+j}} \right\} \quad (2.7)$$

converges absolutely and locally normally on $\operatorname{Re} u > -K$. It is holomorphic there, and

$$\boxed{R_K(u; c) = G(u) - \sum_{j=0}^{K-1} A_j(u; c)\zeta(1+u+j, c).} \quad (2.8)$$

Every apparent pole of the right side inside this half-plane is removable. Arbitrary fixed-order spectral and parameter derivatives can be taken termwise in the bracketed series.

Proof. On a compact subset with $\operatorname{Re} u \geq -K+\delta$, Lemma 2.1 bounds the bracket by a constant times $(n+c)^{-1-\delta}$; derivatives add only a fixed logarithmic factor. These majorants are summable. The Weierstrass theorem and the differentiated version of the same estimate prove normal convergence and holomorphy. In $\operatorname{Re} u > 0$, both separate series converge and (2.8) follows from (2.2) and the defining Dirichlet series for Hurwitz zeta. The identity theorem gives (2.8) on the larger domain, with all singularities removable because its left side is holomorphic. $\square$

Remark 2.3 (Ordinary sums, not unassigned divergent sums). At $u = -N$, $N < K$, the two components inside the braces in (2.7) usually have divergent sums separately. The theorem evaluates the single absolutely convergent sum of their difference. It does not assign values to either divergent series or authorize rearranging the separate pieces. The value depends on the stated reference shift c ; Section 6 gives its exact dependence.

Increasing the number of counterterms gives the exact relation

$$R_{K+1}(u; c) = R_K(u; c) - A_K(u; c)\zeta(1 + u + K, c), \quad \operatorname{Re} u > -K. \quad (2.9)$$

Thus additional subtractions improve summability without creating a new unknown constant. The meromorphic continuation of this relation remains valid wherever its combined terms are defined.

3 The all-integer resonant evaluation

3.1 Reference shifts and a coefficient differential identity

The coefficient polynomials have more structure than their recursive definition initially suggests.

Lemma 3.1 (Exact transport in the reference shift). *For arbitrary reference shifts c, d ,*

$$A_j(u; c) = \sum_{l=0}^j A_l(u; d) \frac{(1 + u + l)_{j-l}}{(j-l)!} (c - d)^{j-l}. \quad (3.1)$$

In particular, A_j has degree at most j in c , and

$$\partial_c A_j(u; c) = (u + j) A_{j-1}(u; c), \quad j \geq 1. \quad (3.2)$$

Proof. In the expansion with reference d , write $n + d = (n + c) - (c - d)$ and expand each factor by the binomial theorem. Uniqueness of asymptotic coefficients gives (3.1). Taking its first-order variation in $c - d$ proves (3.2). Since these are polynomial identities, establishing them for positive references proves their algebraic form without further restrictions. $\square$

Lemma 3.2 (Coefficient differential law). *For $j \geq 1$,*

$$(\partial_a + \partial_b - \partial_u) A_j = -(u + j) A_{j-1} + \sum_{k=1}^j \zeta(1 - k, c) A_{j-k}. \quad (3.3)$$

Proof. Treating n temporarily as a positive real variable, logarithmic differentiation of (2.1) gives the exact identity

$$(\partial_a + \partial_b - \partial_u) r_n(u) = \partial_n r_n(u) + \psi(n + 1) r_n(u). \quad (3.4)$$

The digamma asymptotic expansion at the same reference is

$$\psi(n + 1) \sim \log(n + c) + \sum_{k \geq 1} \frac{\zeta(1 - k, c)}{(n + c)^k}. \quad (3.5)$$

Indeed, its coefficients are $-B_k(c)/k = \zeta(1 - k, c)$, using Bernoulli reflection. Substitute the finite expansions into (3.4). The logarithmic terms cancel because $-\partial_u(n + c)^{-1-u}$ is $\log(n + c)(n + c)^{-1-u}$. At inverse-power order j , $\partial_n r_n$ contributes $-(u + j) A_{j-1}$ and the nonlogarithmic digamma terms contribute the sum in (3.3). Uniqueness of the asymptotic expansion completes the proof. $\square$

3.2 Residue cancellation at every nonpositive integer

Set

$$\rho_N(a, b) = \frac{(-1)^N (1-a)_N (1-b)_N}{N!}. \quad (3.6)$$

This is a polynomial, including when one of the factors vanishes.

Proposition 3.3 (Two exact resonant coefficient identities). *For every $N \geq 0$,*

$$A_N(-N; c) = \rho_N(a, b), \quad (3.7)$$

$$\partial_u A_N(-N; c) + \sum_{j=0}^{N-1} A_j(-N; c) \zeta(1 - N + j, c) = (\partial_a + \partial_b) \rho_N(a, b). \quad (3.8)$$

The sum in the second identity is empty for $N = 0$.

Proof. Take $K = N + 1$ in Theorem 2.2. The only zeta factor with a pole at $u = -N$ has index $j = N$ and residue $A_N(-N; c)$. On the other hand, $\text{Res}_{u=-N} G(u) = \rho_N$, by the gamma residue formula and $\Gamma(a)/\Gamma(a - N) = (a - N)_N = (-1)^N (1 - a)_N$. Since R_{N+1} is holomorphic, its residue vanishes, proving (3.7). For generic parameters this is a direct residue calculation; continuity gives the polynomial statement at degeneracies.

For $N \geq 1$, put $j = N$, $u = -N$ in (3.3). The factor $u + j$ vanishes. Use (3.7) for the a, b derivatives and rearrange. This proves (3.8). For $N = 0$, both sides are zero because $A_0 = \rho_0 = 1$. $\square$

The second identity is an explicit finite Bernoulli-polynomial identity, not merely a residue statement. It is the cancellation needed to turn a long finite-part formula into a short evaluation.

Theorem 3.4 (Closed evaluation at every resonance). *For positive a, b, c and every $N \geq 0$,*

$$\begin{aligned} \sum_{n=0}^{\infty} \left\{ \frac{\Gamma(n+a)\Gamma(n+b)}{\Gamma(n+1)\Gamma(n+a+b-N)} - \sum_{j=0}^N A_j(-N; c) (n+c)^{N-1-j} \right\} \\ = \boxed{\rho_N(a, b) (H_N - \gamma - \psi(a) - \psi(b) + \psi(c))}. \end{aligned} \quad (3.9)$$

The series is absolutely convergent, including at positive integer parameters for which $\rho_N = 0$.

Proof. The case of nondegenerate parameters is enough initially. Put $u = -N + t$. The gamma Laurent expansion gives

$$G(-N + t) = \frac{\rho_N}{t} + \rho_N \{ \psi(N+1) - \psi(a-N) - \psi(b-N) \} + O(t).$$

Writing $A_N(-N + t; c) = \rho_N + t \partial_u A_N(-N; c) + O(t^2)$ and using (1.1), its matching zeta product has constant term $\partial_u A_N(-N; c) - \rho_N \psi(c)$. Consequently (2.8) gives

$$\begin{aligned} R_{N+1}(-N; c) &= \rho_N \{ \psi(N+1) - \psi(a-N) - \psi(b-N) + \psi(c) \} \\ &\quad - \partial_u A_N(-N; c) - \sum_{j=0}^{N-1} A_j(-N; c) \zeta(1 - N + j, c). \end{aligned}$$

The last line is $-(\partial_a + \partial_b) \rho_N$ by (3.8). Also

$$\frac{\partial_a \rho_N}{\rho_N} = \psi(a) - \psi(a-N), \quad \frac{\partial_b \rho_N}{\rho_N} = \psi(b) - \psi(b-N).$$

Substitution cancels the shifted digammas, leaving $\rho_N\{\psi(N+1) - \psi(a) - \psi(b) + \psi(c)\}$. Finally $\psi(N+1) = H_N - \gamma$. Both the normally convergent bracketed sum and this last expression are continuous, indeed locally holomorphic, in the positive parameters. The identity therefore extends across all zeros of ρ_N . $\square$

Remark 3.5 (What is classical, and what the proof adds). The value theorem is consistent with the classical hypergeometric constant-term analysis discussed in Section 1. The proof above explains its simplification entirely within the fixed-shift coefficient algebra, supplies an exact recurrence for every subtraction, and remains valid at reciprocal-gamma degeneracies. The next section retains the entire Taylor expansion rather than only its constant term.

4 All resonant spectral jets and Stieltjes constants

4.1 An analytic gamma factor without singular intermediate terms

The direct Laurent expansion at $-N$ can be awkward when $a - N$ or $b - N$ is a nonpositive integer. The following exact factorization removes that problem. Define

$$E_0(t) = \frac{\Gamma(a)\Gamma(b)\Gamma(1+t)}{\Gamma(a+t)\Gamma(b+t)}, \quad (4.1)$$

$$E_N(t) = E_0(t) \prod_{r=1}^N \frac{(a+t-r)(b+t-r)}{t-r}. \quad (4.2)$$

Every denominator $t - r$ is nonzero near $t = 0$, and the formula is valid also when numerator factors vanish. The gamma functional equation gives

$$E_N(t) = tG(-N+t), \quad E_N(0) = \rho_N. \quad (4.3)$$

Write

$$e_q = [t^q]E_N(t), \quad a_{j,q} = [t^q]A_j(-N+t; c).$$

The coefficients $a_{j,q}$ are finite polynomial data. Those of E_0 are Bell polynomials in ordinary polygamma values at positive arguments, and (4.2) then uses only finite products and divisions by nonvanishing linear factors.

Theorem 4.1 (All-order resonant Stieltjes formula). *Let $K > N$ and $m \geq 0$. Define $S_{N,m}^{(K)}(c) = [t^m]R_K(-N+t; c)$. Then*

$$\begin{aligned} S_{N,m}^{(K)}(c) &= e_{m+1} - a_{N,m+1} - \sum_{h=0}^m a_{N,m-h} \frac{(-1)^h \gamma_h(c)}{h!} \\ &\quad - \sum_{\substack{0 \leq j < K \\ j \neq N}} \sum_{q=0}^m a_{j,q} \frac{\zeta^{[m-q]}(1-N+j, c)}{(m-q)!}. \end{aligned} \quad (4.4)$$

Its left side is the absolutely convergent ordinary series

$$\sum_{n=0}^{\infty} [t^m] \left\{ r_n(-N+t) - \sum_{j=0}^{K-1} A_j(-N+t; c) (n+c)^{N-1-j} e^{-t \log(n+c)} \right\}. \quad (4.5)$$

For the smallest uniform choice $K = N+1$, the only zeta derivatives in the last line of (4.4) occur at $0, -1, \dots, 1-N$.

Proof. Theorem 2.2 permits coefficient extraction from (2.8) and termwise from (2.7). The gamma term is $E_N(t)/t$, so its coefficient of t^m is e_{m+1} . The singular counterterm is

$$A_N(-N+t; c) \left\{ \frac{1}{t} + \sum_{h \geq 0} \frac{(-1)^h \gamma_h(c)}{h!} t^h \right\}.$$

Its coefficient is $a_{N,m+1}$ plus the first sum in (4.4). All other zeta factors are analytic at $t = 0$, and the Cauchy product gives the remaining finite double sum. The coefficient of t^{-1} cancels by (3.7). This proves the formula, with no appeal to a numerical limiting procedure. $\square$

Corollary 4.2 (Finite closure of each jet). *At fixed N, m and $K = N + 1$, the resonant jet belongs to the algebra generated by gamma/polygamma data at $a, b, 1$, the constants $\gamma_0(c), \dots, \gamma_m(c)$, and $\zeta^{[q]}(-r, c)$ for $0 \leq r \leq N - 1$, $0 \leq q \leq m$, with polynomial coefficients in a, b, c and rational numerical factors. Only finitely many members of these lists occur in any one evaluation.*

This is a proved representation statement, not a claim of algebraic independence or minimality. Lerch’s identity $\zeta^{[1]}(0, c) = \log \Gamma(c) - \frac{1}{2} \log(2\pi)$ already reduces some of the listed terms [3, 25.11.18]. Negative spectral arguments connect the formula to the integrated Hurwitz-zeta/negative-polygamma framework in the canonical manuscript and in the classical work of Espinosa and Moll [6].

4.2 Explicit Bell formula at zero excess

Set

$$\begin{aligned} L_1 &= -\gamma - \psi(a) - \psi(b), \\ L_k &= (-1)^k (k-1)! \zeta(k) - \psi^{(k-1)}(a) - \psi^{(k-1)}(b), \quad k \geq 2. \end{aligned} \quad (4.6)$$

Then $\log E_0(t) = \sum_{k \geq 1} L_k t^k / k!$. For $N = 0$, $K = 1$, Theorem 4.1 becomes the all-order identity

$$\begin{aligned} \sum_{n=0}^{\infty} \left\{ r_n(0) \mathcal{B}_m(-\psi(n+a+b), -\psi'(n+a+b), \dots, -\psi^{(m-1)}(n+a+b)) - \frac{(-\log(n+c))^m}{n+c} \right\} \\ = \frac{\mathcal{B}_{m+1}(L_1, \dots, L_{m+1})}{m+1} - (-1)^m \gamma_m(c). \end{aligned} \quad (4.7)$$

For $m = 0$ the Bell polynomial in the summand is one. The expression is a derivative formula; its right side has been multiplied by $m!$ relative to (4.4).

For instance, the first two nontrivial levels are

$$\sum_{n \geq 0} \left\{ -r_n(0) \psi(n+a+b) + \frac{\log(n+c)}{n+c} \right\} = \frac{L_1^2 + L_2}{2} + \gamma_1(c), \quad (4.8)$$

$$\sum_{n \geq 0} \left\{ r_n(0) (\psi(n+a+b)^2 - \psi'(n+a+b)) - \frac{\log^2(n+c)}{n+c} \right\} = \frac{L_1^3 + 3L_1L_2 + L_3}{3} - \gamma_2(c). \quad (4.9)$$

The paired counterterms are essential; the separate logarithmic sums are divergent.

4.3 Terms that vanish at resonance can revive under differentiation

For $k \geq 0$, the exact local reciprocal-gamma identity is

$$\frac{1}{\Gamma(-k+t)} = (-1)^k k! t \frac{\prod_{r=1}^k (1-t/r)}{\Gamma(1+t)}. \quad (4.10)$$

It follows directly by applying the gamma functional equation $k+1$ times. In particular,

$$\frac{1}{\Gamma(-k+t)} = (-1)^k k! t + O(t^2). \quad (4.11)$$

Thus a zero summand at $t=0$ is not a zero Taylor series.

For $a = b = \frac{1}{2}$, define

$$w_n = \pi \frac{\binom{2n}{n}^2}{16^n}. \quad (4.12)$$

At $u = -N$, $r_n(-N) = w_n n^N$, so every $n < N$ term vanishes. For $n < N$, put $k = N - n - 1$. Formula (4.10) gives

$$\begin{aligned} r_n(-N+t) &= w_n n! (-1)^k k! t \\ &\times \exp \left\{ (\gamma - H_k) t - \sum_{r \geq 2} \frac{(-1)^r \zeta(r) + H_k^{(r)}}{r} t^r \right\}. \end{aligned} \quad (4.13)$$

This produces every missing finite-head coefficient explicitly. The simplest case is

$$r_0(-1) = 0, \quad [t]r_0(-1+t) = \pi. \quad (4.14)$$

Discarding this term after setting $u = -1$ and then differentiating would change the first-jet evaluation by exactly π . This is a concrete guard against an invalid extension procedure, not an allegation that the canonical manuscript already contains that mistake.

5 Exact harmonic-number and binomial families

5.1 Every falling-factorial moment of the binomial-square weight

For the weights w_n in (4.12), Theorem 3.4 gives

$$\sum_{n \geq 0} \left\{ w_n n^N - \sum_{j=0}^N A_j(-N; 1) (n+1)^{N-1-j} \right\} = \frac{(-1)^N (\frac{1}{2})_N^2}{N!} (H_N + 4 \log 2). \quad (5.1)$$

Here and throughout this subsection A_j is specialized to $a = b = \frac{1}{2}$. Three low-order examples are

$$\sum_{n \geq 0} \left(w_n - \frac{1}{n+1} \right) = 4 \log 2, \quad (5.2)$$

$$\sum_{n \geq 0} \left(n w_n - 1 + \frac{1}{4(n+1)} \right) = -\log 2 - \frac{1}{4}, \quad (5.3)$$

$$\sum_{n \geq 0} \left(n(n-1) w_n - n + \frac{5}{4} - \frac{9}{32(n+1)} \right) = \frac{9}{8} \log 2 + \frac{27}{64}. \quad (5.4)$$

The $n = 0$ terms in these ordinary sums must be retained. The recurrence (2.4) generates the counterterms at every order. The accompanying JSON table records them through $N = 10$.

N	ρ_N	$\rho_N H_N$
0	1	0
1	$-1/4$	$-1/4$
2	$9/32$	$27/64$
3	$-75/128$	$-275/256$
4	$3675/2048$	$30625/8192$
5	$-59535/8192$	$-543753/32768$

Every value is $4\rho_N \log 2 + \rho_N H_N$; the growing counterterm polynomials do not introduce an additional unevaluated constant.

5.2 A directly evaluated Stieltjes–harmonic sum

At $a = b = \frac{1}{2}$, the Bell parameters in (4.6) are

$$L_1 = \gamma + 4 \log 2, \quad L_2 = -5\zeta(2), \quad L_3 = 26\zeta(3). \quad (5.5)$$

Using $\psi(n+1) = H_n - \gamma$ in (4.8), and then (5.2), gives

$$\sum_{n \geq 0} \left\{ w_n H_n - \frac{\log(n+1) + \gamma}{n+1} \right\} = \frac{5}{2}\zeta(2) - 8 \log^2 2 - \frac{\gamma^2}{2} - \gamma_1. \quad (5.6)$$

This is an exact convergent series representation, not a relation inferred from a numerical search. All higher analogues follow from (4.7); for example (4.9) with (5.5) expresses a quadratic harmonic combination using γ_2 .

The first negative-resonance jet is also explicit. With $A = \gamma + 4 \log 2$,

$$\begin{aligned} \pi - \frac{3}{2} + \sum_{n=1}^{\infty} \left\{ n w_n (\gamma - H_{n-1}) + \log(n+1) - \frac{\frac{3}{2} + \frac{1}{4} \log(n+1)}{n+1} \right\} \\ = -\frac{(A-3)^2}{8} + \frac{5}{8}\zeta(2) + \frac{11}{8} - \frac{3}{2}\gamma - \frac{1}{4}\gamma_1 + \frac{1}{2}\log(2\pi). \end{aligned} \quad (5.7)$$

The initial π is precisely the revived zero term of (4.14). Formula (4.4) proves the evaluation; the separately executed series diagnostic agrees with this displayed specialization.

5.3 Complete finite harmonic numbers: an all-order family

Define the complete harmonic numbers by

$$h_n^{(m)} = \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} \frac{1}{j_1 \cdots j_m}, \quad h_n^{(0)} = 1.$$

For $m > 0$, $h_0^{(m)} = 0$. In particular $h_n^{(1)} = H_n$ and $h_n^{(2)} = (H_n^2 + H_n^{(2)})/2$.

Proposition 5.1 (Complete harmonic binomial differences). *For every $m \geq 0$,*

$$(-1)^m \sum_{n \geq 0} \left\{ w_n h_n^{(m)} - \frac{h_{n+1}^{(m)}}{n+1} \right\} = [t^{m+1}] \frac{\pi \Gamma(1+t)^2}{\Gamma(\frac{1}{2}+t)^2}. \quad (5.8)$$

Equivalently the right side is $\mathcal{B}_{m+1}(\lambda_1, \dots, \lambda_{m+1})/(m+1)!$, where

$$\lambda_1 = 4 \log 2, \quad \lambda_k = 2(-1)^k (k-1)! (2-2^k) \zeta(k) \quad (k \geq 2). \quad (5.9)$$

Proof. Multiplication of the half-parameter summand by $\Gamma(1+t)$ gives

$$\Gamma(1+t) r_n(t) = w_n \prod_{j=1}^n (1+t/j)^{-1}.$$

For $a = b = 1$, the analogous expression is $(n+1)^{-1} \prod_{j=1}^{n+1} (1+t/j)^{-1}$. Their coefficient difference is $(-1)^m$ times the summand in (5.8). The leading asymptotic coefficient of both gamma-ratio sequences is one, independent of a, b , so their difference and all fixed jets are normally summable near $t = 0$. For $\operatorname{Re} t > 0$, Gauss summation gives the difference of sums as

$$\frac{1}{t} \left\{ \frac{\pi \Gamma(1+t)^2}{\Gamma(\frac{1}{2}+t)^2} - 1 \right\}.$$

Analytic continuation to zero and coefficient extraction prove (5.8). Logarithmic differentiation of the gamma quotient proves (5.9). $\square$

For example,

$$\sum_{n \geq 0} \left\{ w_n H_n - \frac{H_{n+1}}{n+1} \right\} = 2\zeta(2) - 8 \log^2 2, \quad (5.10)$$

$$\sum_{n \geq 0} \left(w_n - \frac{1}{n+1} \right) H_n = 3\zeta(2) - 8 \log^2 2, \quad (5.11)$$

$$\sum_{n \geq 0} \left\{ w_n \frac{H_n^2 + H_n^{(2)}}{2} - \frac{H_{n+1}^2 + H_{n+1}^{(2)}}{2(n+1)} \right\} = \frac{32}{3} \log^3 2 - 8 \log 2 \zeta(2) + 4\zeta(3). \quad (5.12)$$

The final coefficient is $4\zeta(3)$: it follows from $\lambda_3 = 24\zeta(3)$ and the Taylor factorial $3!$. This explicit normalization is useful in transcribing higher formulas.

5.4 A balanced parameter identity and complete harmonic tails

An especially simple all-weight result follows by varying the two numerator parameters in opposite directions.

Proposition 5.2 (Balanced gamma-difference identity). *Let $a, b > 0$ and $-a < \operatorname{Re} t < b$. Then*

$$\begin{aligned} \sum_{n \geq 0} \frac{\Gamma(n+a)\Gamma(n+b)}{\Gamma(n+1)\Gamma(n+a+b)} \left\{ \frac{\Gamma(n+a+t)\Gamma(n+b-t)}{\Gamma(n+a)\Gamma(n+b)} - 1 \right\} \\ = \psi(a) + \psi(b) - \psi(a+t) - \psi(b-t). \end{aligned} \quad (5.13)$$

The series is absolutely locally normally convergent in this strip.

Proof. Subtract (3.9) for $N = 0$ at the parameter pairs $(a + t, b - t)$ and (a, b) , keeping the same c . The denominator $n + a + b$ is unchanged and the $1/(n + c)$ counterterms cancel. The numerator parameter sum is fixed, so the leading asymptotic coefficient remains one; the difference is $O(n^{-2})$, uniformly on compact subsets of the strip. Holomorphy extends the initially real-parameter identity throughout the strip. $\square$

For $x > 0$ define the complete harmonic tails

$$T_{n,k}(x) = \sum_{n \leq j_1 \leq \dots \leq j_k} \frac{1}{(j_1 + x)^2 \cdots (j_k + x)^2}, \quad k \geq 1. \quad (5.14)$$

These are finite real numbers, although the nested tails are infinite. They are complete homogeneous polynomials in the tail power sums:

$$\sum_{k \geq 0} T_{n,k}(x) y^k = \exp \left(\sum_{r \geq 1} \frac{\zeta(2r, n + x)}{r} y^r \right), \quad T_{n,0} = 1. \quad (5.15)$$

In particular

$$T_{n,1} = \zeta(2, n + x), \quad T_{n,2} = \frac{1}{2} \{ \zeta(2, n + x)^2 + \zeta(4, n + x) \}.$$

Theorem 5.3 (All-weight gamma-weighted harmonic-tail collapse). *For $x > 0$ and every integer $k \geq 1$,*

$$\boxed{\sum_{n=0}^{\infty} \frac{\Gamma(n + x)^2}{\Gamma(n + 1)\Gamma(n + 2x)} T_{n,k}(x) = 2\zeta(2k + 1, x).} \quad (5.16)$$

The sum is absolutely convergent.

Proof. Put $a = b = x$ in (5.13). For $|t| < x$, the gamma-product expansion is

$$\begin{aligned} \frac{\Gamma(n + x + t)\Gamma(n + x - t)}{\Gamma(n + x)^2} &= \exp \left(\sum_{r \geq 1} \frac{\zeta(2r, n + x)}{r} t^{2r} \right) \\ &= \sum_{k \geq 0} T_{n,k}(x) t^{2k}. \end{aligned}$$

The first equality follows by adding the Taylor expansions of the two log-gamma functions, using $\psi^{(2r-1)}(y) = (2r - 1)!\zeta(2r, y)$. It also follows from the convergent product $\prod_{j=n}^{\infty} (1 - t^2/(j + x)^2)^{-1}$. The right side of (5.13) expands as

$$2\psi(x) - \psi(x + t) - \psi(x - t) = 2 \sum_{k \geq 1} \zeta(2k + 1, x) t^{2k}.$$

Local normal convergence in Proposition 5.2 permits coefficient extraction, proving (5.16). Alternatively, for real $0 < t < x$ all tail coefficients are nonnegative, and Tonelli followed by uniqueness of the power series gives the same conclusion. Absolute convergence is also immediate from $T_{n,k}(x) = O(n^{-k})$ and the weight $O(n^{-1})$, or from the coefficient theorem itself. $\square$

At $x = 1$, the left side is the familiar starred multiple sum $\zeta^*(\{2\}^k, 1)$; the theorem includes its value $2\zeta(2k + 1)$. We do not claim priority for that specialization. At $x = \frac{1}{2}$, set

$$U_{n,r} = \zeta(2r, n + \tfrac{1}{2}) = (2^{2r} - 1)\zeta(2r) - 2^{2r} H_{2n}^{(2r)} + H_n^{(2r)}. \quad (5.17)$$

Then the all-weight result becomes

$$\sum_{n \geq 0} w_n T_{n,k}(\tfrac{1}{2}) = 2(2^{2k+1} - 1)\zeta(2k + 1). \quad (5.18)$$

Three explicit evaluations are

$$\sum_{n \geq 0} w_n U_{n,1} = 14\zeta(3), \quad (5.19)$$

$$\sum_{n \geq 0} w_n \frac{U_{n,1}^2 + U_{n,2}}{2} = 62\zeta(5), \quad (5.20)$$

$$\sum_{n \geq 0} w_n \frac{U_{n,1}^3 + 3U_{n,1}U_{n,2} + 2U_{n,3}}{6} = 254\zeta(7). \quad (5.21)$$

The combinations in (5.17) are small harmonic *tails*; one must not split their zeta and harmonic parts into divergent weighted sums.

5.5 Other rational-parameter exact families

Take $b = 1 - a$, $c = 1$, and define $v_n(a) = \Gamma(n + a)\Gamma(n + 1 - a)/\Gamma(n + 1)^2$. Then $r_n(-N) = v_n(a)n^{\underline{N}}$, and the closed value is

$$\frac{(-1)^N (a)_N (1 - a)_N}{N!} (H_N + C_a), \quad C_a = -2\gamma - \psi(a) - \psi(1 - a), \quad (5.22)$$

after the same explicit polynomial/harmonic subtraction. Standard rational digamma identities give

$$C_{1/2} = 4 \log 2, \quad C_{1/3} = 3 \log 3, \quad C_{1/4} = 6 \log 2, \quad C_{1/6} = 4 \log 2 + 3 \log 3.$$

For example,

$$v_n(1/3) = \frac{2\pi}{\sqrt{3}} \frac{(3n)!}{27^n (n!)^3}, \quad v_n(1/4) = \pi\sqrt{2} \frac{(4n)!}{64^n (2n)!(n!)^2}.$$

Thus every falling-factorial moment in these two factorial-ratio families has an exact subtracted evaluation in a single logarithm and rational numbers. The general jet theorem supplies their Stieltjes analogues without a new summation argument.

6 An exact telescoping antiderivative calculus

Theorem 6.1 (Reference-parameter telescoping). *For $K \geq 1$ and $\operatorname{Re} u > -K$,*

$$\boxed{\partial_c R_K(u; c) = (u + K) A_{K-1}(u; c) \zeta(1 + u + K, c).} \quad (6.1)$$

Consequently an exact primitive is

$$\int A_{K-1}(u; c) \zeta(1 + u + K, c) \, dc = -\frac{1}{u + K} \sum_{j=0}^{K-1} A_j(u; c) \zeta(1 + u + j, c) + C. \quad (6.2)$$

At a removable resonance, the finite sum on the right is evaluated as a combined Laurent expression; any residual pole is independent of c and can be absorbed into the integration constant before taking the limit. Equivalently, $R_K(u; c)/(u + K)$ is a primitive that is holomorphic throughout the stated half-plane.

Proof. Differentiate the finite counterterm sum in (2.8), using (3.2) and $\partial_c \zeta(s, c) = -s\zeta(s+1, c)$. For $j \geq 1$ the derivative of the coefficient contributes $(u+j)A_{j-1}\zeta(1+u+j, c)$, which cancels the derivative of the preceding zeta factor. Only $-(u+K)A_{K-1}\zeta(1+u+K, c)$ remains from differentiating the sum. The c -independent $G(u)$ contributes zero, proving (6.1). Division by $u+K$ is harmless on $\operatorname{Re} u > -K$ and gives the primitive. The last statement explains how to read its meromorphic finite-sum form at a resonance. $\square$

For $K = N + 1$ and $u = -N + t$, the all-jet primitive law is particularly simple:

$$\partial_c S_{N,m}^{(N+1)}(c) = [t^m] \{ (1+t)A_N(-N+t; c)\zeta(2+t, c) \}. \quad (6.3)$$

Thus the finite Stieltjes expression in (4.4) is an explicitly evaluated antiderivative of a polynomial-weighted combination of spectral zeta derivatives at 2. This is stronger than merely leaving the antiderivative as an unevaluated integral.

The first level recovers

$$\gamma'_1(c) = \zeta(2, c) + \zeta^{[1]}(2, c), \quad (6.4)$$

by differentiating (1.1) in c . For $N = 1$, let $\rho = -(1-a)(1-b)$ and $e_2 = [t^2]E_1(t)$. Since

$$A_1(-1+t; c) = \rho + (c-a-b+\tfrac{3}{2})t - \tfrac{1}{2}t^2,$$

Theorem 4.1 becomes

$$S_{1,1}^{(2)}(c) = e_2 + \frac{1}{2} + \frac{1}{2} \log(2\pi) - \log \Gamma(c) + (c-a-b+\tfrac{3}{2})\psi(c) + \rho\gamma_1(c). \quad (6.5)$$

Taking the c derivative yields the concrete integral identity

$$\begin{aligned} \int \{ (c-a-b+\tfrac{3}{2}+\rho)\zeta(2, c) + \rho\zeta^{[1]}(2, c) \} dc \\ = -\log \Gamma(c) + (c-a-b+\tfrac{3}{2})\psi(c) + \rho\gamma_1(c) + C. \end{aligned} \quad (6.6)$$

For $a = b = \frac{1}{2}$ this reads

$$\int \left\{ (c+\tfrac{1}{4})\psi'(c) - \tfrac{1}{4}\zeta^{[1]}(2, c) \right\} dc = -\log \Gamma(c) + (c+\tfrac{1}{2})\psi(c) - \tfrac{1}{4}\gamma_1(c) + C. \quad (6.7)$$

The all-order theorem gives the analogous formulas with higher spectral jets, rather than requiring separate integration-by-parts calculations.

At zeroth order, (6.3) says $\partial_c R_{N+1}(-N; c) = \rho_N \zeta(2, c)$. Integrating from c to d gives the exact reference-change rule

$$R_{N+1}(-N; d) - R_{N+1}(-N; c) = \rho_N \{ \psi(d) - \psi(c) \}. \quad (6.8)$$

This makes the subtraction convention fully visible. The finite constant is not invariant under replacing $(n+c)^{-1}$ by an unspecified harmonic reference term.

7 Polylogarithmic endpoint identities and z -primitives

7.1 Regularized hypergeometric notation

For $|z| < 1$ define

$$F(u, z) = \sum_{n \geq 0} r_n(u) z^n = \Gamma(a)\Gamma(b) {}_2F_1(a, b; a+b+u; z), \quad (7.1)$$

where ${}_2\mathbf{F}_1(a, b; d; z) = {}_2F_1(a, b; d; z)/\Gamma(d)$ is the *regularized* function, defined by its coefficient series. This convention is indispensable when d is a nonpositive integer: separate floating-point evaluation of an ordinary hypergeometric pole and a reciprocal-gamma zero is not the definition.

The Lerch transcendent satisfies

$$\Phi(z, s, c) = \sum_{n \geq 0} \frac{z^n}{(n+c)^s}, \quad z\Phi(z, s, 1) = \text{Li}_s(z), \quad (7.2)$$

initially on $|z| < 1$ [4].

Theorem 7.1 (Polylogarithmic endpoint subtraction, with all jets). *Let $K \geq 1$. The combined function*

$$F_K(u, z; c) = F(u, z) - \sum_{j=0}^{K-1} A_j(u; c) \Phi(z, 1+u+j, c) \quad (7.3)$$

has the absolutely convergent coefficient representation

$$F_K(u, z; c) = \sum_{n \geq 0} \left\{ r_n(u) - \sum_{j=0}^{K-1} \frac{A_j(u; c)}{(n+c)^{1+u+j}} \right\} z^n$$

for every $|z| \leq 1$ and $\text{Re } u > -K$. It is continuous in z on the closed disk and holomorphic in u . The convergence and every fixed-order u derivative are uniform on the closed disk and compact subsets of that half-plane. In particular,

$$\lim_{z \rightarrow 1, |z| \leq 1} [t^m] F_K(-N+t, z; c) = S_{N,m}^{(K)}(c), \quad N < K. \quad (7.4)$$

For $c = 1$ this is an endpoint identity with finitely many ordinary polylogarithms and their spectral derivatives:

$$zF_K(u, z; 1) = zF(u, z) - \sum_{j=0}^{K-1} A_j(u; 1) \text{Li}_{1+u+j}(z). \quad (7.5)$$

Proof. The coefficient bracket is the one in Theorem 2.2. The same summable majorant works for every $|z| \leq 1$, since $|z^n| \leq 1$. The Weierstrass test, also after a fixed number of spectral derivatives, proves uniform convergence, continuity, and interchange of endpoint limits with coefficient extraction. Equations (7.3) and (7.5) first hold inside the disk by absolutely convergent power series, and then represent the claimed continuous extension of the *combined* expression. $\square$

Remark 7.2 (The separate pieces can still be singular). The theorem does not assert that each Lerch or polylogarithm term has a finite endpoint separately. Nor does it rely on identifying two independently regularized boundary values. The coefficient differences already form an ordinary uniformly convergent series, which fixes the limit uniquely. The spectral-derivative formula is therefore meaningful at negative integer polylogarithm orders and at order one alike.

7.2 Explicit endpoint examples

In the half-parameter case, write $\mathcal{F}(z) = \sum_{n \geq 0} w_n z^n = \pi {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; z)$ and $\theta = z\partial_z$. For $N \geq 0$, the resonant series is $F(-N, z) = \theta(\theta-1) \cdots (\theta-N+1)\mathcal{F}(z)$. For instance,

$$\lim_{z \rightarrow 1^-} \{z\mathcal{F}(z) - \text{Li}_1(z)\} = 4 \log 2, \quad (7.6)$$

$$\lim_{z \rightarrow 1^-} \{z\theta\mathcal{F}(z) - \text{Li}_0(z) + \frac{1}{4} \text{Li}_1(z)\} = -\log 2 - \frac{1}{4}, \quad (7.7)$$

$$\lim_{z \rightarrow 1^-} \{z\theta(\theta-1)\mathcal{F}(z) - \text{Li}_{-1}(z) + \frac{9}{4} \text{Li}_0(z) - \frac{9}{32} \text{Li}_1(z)\} = \frac{9}{8} \log 2 + \frac{27}{64}. \quad (7.8)$$

These are ordinary limits of the displayed expressions. They are instances of the all- N formula, not numerical extrapolations.

The first order-derivative example is

$$\lim_{z \rightarrow 1^-} \{z \partial_u F(u, z)|_{u=0} - \partial_s \text{Li}_s(z)|_{s=1}\} = \frac{(\gamma + 4 \log 2)^2 - 5\zeta(2)}{2} + \gamma_1. \quad (7.9)$$

All higher orders and all negative resonances follow by the explicit finite formula (4.4). This is the direct connection between the polylogarithmic order calculus and the Stieltjes layer in this construction.

7.3 Exact Euler derivatives and normalized antiderivatives

The operator θ multiplies a coefficient of z^n by n . After q applications, the endpoint majorant becomes $O(n^{q-1-\text{Re } u-K} \log^m n)$. Thus the same endpoint statement for $\theta^q F_K$ is valid on the smaller half-plane

$$\text{Re } u > q - K. \quad (7.10)$$

One cannot differentiate the boundary formula arbitrarily often while keeping the original number of counterterms.

For $c > 0$, define the zero-basepoint normalized primitive

$$(\mathcal{I}_c f)(z) = z^{-c} \int_0^z t^{c-1} f(t) dt. \quad (7.11)$$

It is first interpreted for real $0 < z < 1$, and then by its analytic power series. Coefficientwise, $\mathcal{I}_c(z^n) = z^n/(n+c)$; hence

$$\mathcal{I}_c^\ell \Phi(z, s, c) = \Phi(z, s + \ell, c), \quad \ell \geq 0. \quad (7.12)$$

For the gamma-ratio term this gives the exact lift

$$\mathcal{I}_c^\ell F(u, z) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b+u)c^{\ell+2}} F_{\ell+1} \left(\begin{matrix} a, b, \underbrace{c, \dots, c}_\ell \\ a+b+u, \underbrace{c+1, \dots, c+1}_\ell \end{matrix}; z \right). \quad (7.13)$$

As in (7.1), reciprocal-gamma regularization is understood at an exceptional lower parameter. The proof is the elementary identity $(c)_n/(c+1)_n = c/(n+c)$ in every coefficient.

Applying (7.12)–(7.13) to (7.3) provides exact antiderivatives of all the endpoint-subtracted families. Spectral differentiation commutes with these primitives inside the disk, and with their endpoint limits whenever the corresponding coefficient majorant is summable. This is an integration ladder, not a claim that arbitrary higher hypergeometric endpoint values reduce to gamma values. Likewise, the generic function $F(u, z)$ is not asserted to be an ordinary polylogarithm. The proved polylogarithmic statement is the finite subtraction and the exact evaluation of its combined endpoint.

8 Corrections, proof status, and reproducible evidence

8.1 A narrowly scoped manuscript wording correction

The pinned source `chapters/07-integration.tex`, lines 41–44 in the inspected file, describes a PSLQ run detecting a relation among supplied constants rather than a relation involving the target.

It then states: “Basis atoms must be $\mathbb{Q}$ -independent” [9]. That is too strong as a requirement on PSLQ. An integer-relation algorithm accepts a vector whose entries may be dependent; finding a dependence is precisely its purpose [7]. The practical difficulty is that a pre-existing relation can have zero coefficient on the target, making that output uninformative for the intended evaluation.

The proposed replacement is:

Known rational dependencies among the supplied constants should be removed or modeled explicitly. Otherwise PSLQ may return a pre-existing relation with zero target coefficient instead of an evaluation of the target.

This is an editorial correction to the inference being drawn, not a change to the recorded special-function value. Related cautions already occur in earlier continuation material; no priority is claimed for noticing it. The package includes a minimal patch tied to the inspected context, but it has not been applied to the repository.

8.2 Three guard conditions for the new calculus

The first guard is the reciprocal-gamma issue of (4.14): evaluating at a zero before differentiating loses nonzero Taylor coefficients. The second is normalization: our jets are coefficients, while (4.7) is a derivative formula. For example the half-parameter values are $L_3 = 26\zeta(3)$ and $\lambda_3 = 24\zeta(3)$, so the cubic coefficient in the normalized gamma quotient is not obtained by dividing an incorrectly shared Bell parameter. The third guard is the combined nature of every subtraction. Neither the harmonic tails in (5.17) nor the Stieltjes counterterms in (4.5) can be split into independently summed divergent pieces.

These are explicitly proved implementation safeguards. Apart from the wording correction just identified, they are not asserted to be newly found flaws in the canonical manuscript or in the uninspected incoming archives.

8.3 Analytic proofs versus finite checks

All theorems in this article have analytic proofs. The proof of normal convergence and coefficient extraction does not use a numerical tolerance. The exact test suite verifies polynomial formulas and selected specializations; the numerical suite is an independent diagnostic of normalization, signs, degeneracies, and transcriptions.

The delivered exact run completed **512 checks**. They include generic polynomial identities through A_4 , the finite reference-change identity, residues and the Bernoulli cancellation at rational parameter pairs through $N = 8$, a second path-jet generator, and the reciprocal-gamma polynomial factor. The central-binomial counterterm table is generated through $N = 10$. The checks use rational arithmetic and symbolic polynomial identities; they are not a proof-assistant formalization.

The delivered numerical run completed **64 diagnostics at 65 decimal working digits**. The primary series evaluation uses 108 actual terms and asymptotic coefficients through $j = 25$. It is repeated with 72 terms and coefficients through $j = 21$. The closed side is evaluated from the finite gamma/Stieltjes expression. The tests include spectral jets through order three at resonances $N = 0, 1, 2, 3$, non-half rational parameters, positive integer degeneracies, the harmonic-tail identities through $k = 3$, interior polylogarithmic identities, and the primitive telescope. The largest recorded scaled discrepancy was

$$1.80742922933 \times 10^{-43}, \quad \text{scaled error} = \frac{|\text{left} - \text{right}|}{\max(1, |\text{right}|)}. \quad (8.1)$$

These are *not rigorous interval enclosures*. In particular, (8.1) is not a certificate of 43 correct decimal places. The analytic identities are proved in the text; the diagnostics verify that independently constructed evaluations agree to the reported precision. Full parameter choices and residuals are in `results/numerical_checks.json`.

8.4 Integration status

The manuscript-ready mathematical components are the subtraction theorem, the coefficient differential law, the all-resonance evaluation, the all-order Stieltjes formula, the harmonic-tail theorem, and the two primitive/endpoint calculi. The detailed sections can be integrated after the canonical Hurwitz-jet material, with a cross-reference from the harmonic-number and polylogarithm sections. Formula labels and the local coefficient convention should be preserved during integration.

Nothing here changes the status of S_6 , S_8 , or any numerical independence question. The package contains sources and suggested integration instructions only. No commit, pull request, or automatic modification of ProveIt has been made.

9 Further research questions and concrete next targets

The following questions concern exact identities and structural reductions. They are not assertions that numerical non-detection proves an obstruction.

1. Classify summation families admitting complete jet closure. For a general gamma-ratio sequence of ${}_{p+1}F_p$ type, the same Bernoulli subtraction produces a normally convergent remainder, but the unsubtracted endpoint need not have a gamma-product evaluation. Determine precisely which classical summation submanifolds (for example, balanced or well-poised parameter constraints) have a meromorphic gamma product whose complete resonant jets yield finite Stieltjes closure. A useful first target is a fixed Dougall-type summation with one excess parameter and a fully explicit residue polynomial. The essential issue is an exact summation law, not merely existence of asymptotic coefficients.

2. Establish a multivariate resonance calculus without singular quotients. Formula (4.2) handles all zeros of ρ_N for the present one-parameter spectral problem. For several simultaneous denominator resonances, develop a Taylor algebra based on entire reciprocal-gamma factors and finite head corrections. Determine whether the resulting constant terms depend on the path of a multivariate limit, and state the normalization before introducing a multivariate finite part. This would prevent the loss of terms illustrated by (4.14).

3. Identify the exact special-value algebra of rational harmonic-tail families. Theorem 5.3 gives a continuous x -parameter extension of a multiple-zeta-star evaluation. At rational x , decompose $\zeta(2k+1, x)$ into cyclotomic or character coordinates, and express the left-hand gamma weights and tails in a compatible alphabet. The task is to obtain short explicit identities at denominators 3, 4, 5, 6, not to claim minimal depth or independence. The half-parameter formulas (5.19)–(5.21) provide a model with no residual multiple constant on the right.

4. Construct mixed-order tail identities. The balanced variation $a = x+t, b = x-t$ generates only even tail power sums $\zeta(2r, n+x)$. Independent variations of a, b generate both parities and mixed Bell polynomials. Determine which prescribed combinations of odd and even tails have

a single-zeta evaluation, which have finite products of zeta values, and which retain nontrivial hypergeometric data. The exact balanced identity (5.13) provides a two-parameter starting point, with all coefficient interchanges already justified.

5. Compress the primitive algebra at higher resonances. Theorem 4.1 permits nonpositive-integer zeta derivatives up to $1 - N$. At small (N, m) these can sometimes be rewritten in log-gamma, Barnes-type, or normalized negative-polygamma coordinates. Produce canonical normalizations and change-of-basis formulas through, say, $N = 4$, $m = 3$, including their rational-argument distribution laws. A representation in a smaller *specified* algebra is a meaningful exact target; a claim that the remaining constants are genuinely new requires an additional independence theorem.

6. Compare the fixed-shift and local- z counterterm bases exactly. The present subtraction uses $(n + c)^{-1-u-j}$ and hence Lerch functions. Classical continuation near $z = 1$ instead uses powers of $1 - z$ and logarithms. Determine finite connection matrices between these two counterterm systems at arbitrary resonance and jet order, keeping every constant term. A complete comparison would sharpen the priority map with the classical partial-sum literature and could explain additional simplifications of (4.4).

7. Separate this calculus from the outstanding Gaussian certificates. The canonical S_6 and revised S_8 conjectures still require exact relations in their stated colored-polylogarithm coordinates. A useful next attempt is to introduce a gamma-ratio parameter into an integral representation of those coordinates and test whether the derivatives land in a finite family controlled by a proved summation identity. Any success must produce an explicit analytic identity or a replayable exact relation certificate. The present endpoint results alone do not supply that missing certificate.

8. Formalize the finite coefficient layer and its analytic interface. The recursion, reference transport, and coefficient differential law are polynomial identities suitable for exact formal verification. The analytic interface is a compact-normal convergence theorem for the subtracted series. Formalizing both layers would turn the present regression tests into a machine-checked dependency chain, but no such formalization is included or claimed here.

10 Conclusion

A single classical gamma-ratio summation, paired with a fixed Hurwitz subtraction basis, gives a surprisingly extensive exact calculus. Its zeroth-order resonances collapse to one digamma expression; its higher spectral coefficients introduce Stieltjes constants in a controlled finite way; balanced numerator variation gives single-zeta evaluations of entire harmonic-tail families. Reference differentiation telescopes, while the generating variable supplies polylogarithmic endpoint subtractions and explicit normalized primitives.

The important distinctions are structural. An ordinary convergent bracketed sum is not an unnormalized finite part. A zero reciprocal-gamma value is not a zero jet. A finite closure formula is not an independence result. With these distinctions retained, the identities are directly usable as a new, proved bridge within the ProveIt special-function programme.

A Implementation formulas and replay instructions

A.1 Exact Taylor coefficients without large symbolic expressions

For rational a, b, c, u_0 , the implementation directly computes $[t^q]\ell_k(u_0 + t; c)$. The constant coefficient is (2.3); for $q \geq 1$ it is

$$[t^q]\ell_k(u_0 + t; c) = \frac{(-1)^{k+2}}{k(k+1)} \binom{k+1}{q} B_{k+1-q}(a + b + u_0 - c), \quad q \leq k+1, \quad (\text{A.1})$$

and zero for larger q . Truncated Cauchy multiplication in (2.4) then produces $[t^q]A_j(u_0 + t; c)$ by rational arithmetic. Only the coefficients actually needed by (4.4) are generated.

An independently parameterized routine follows the path $(a + \alpha t, b + \beta t, u_0 + \eta t)$. For $q \geq 1$, its coefficient of ℓ_k is

$$\begin{aligned} \frac{(-1)^{k+1}}{k(k+1)} \binom{k+1}{q} \{ \alpha^q B_{k+1-q}(a - c) + \beta^q B_{k+1-q}(b - c) \\ - (\alpha + \beta + \eta)^q B_{k+1-q}(a + b + u_0 - c) \}. \end{aligned} \quad (\text{A.2})$$

It reduces to (A.1) when $(\alpha, \beta, \eta) = (0, 0, 1)$. The balanced harmonic-tail computation uses $(1, -1, 0)$ and extracts the coefficient of t^{2k} . This avoids evaluating large symbolic gamma derivatives at a singular lower parameter.

A.2 Independent series-side diagnostics

For the spectral jet at $u = -N$, the diagnostic computes the first M terms of (4.5) directly. Its tail is approximated by

$$\sum_{j=K}^{J-1} \sum_{q=0}^m a_{j,q} \frac{\zeta^{[m-q]}(1 - N + j, M + c)}{(m - q)!}. \quad (\text{A.3})$$

These zeta arguments are > 1 for the minimal choice $K = N + 1$; there is no Stieltjes constant or Gauss closed sum on this side of the comparison. The calculation is repeated with different (M, J) . Because the omitted asymptotic remainder is not enclosed by interval arithmetic in the supplied script, the result remains a diagnostic rather than a certified error bound.

For reciprocal-gamma zeros the direct summand jet is computed from (4.10). Elsewhere it is obtained by the exponential recurrence for the Taylor coefficients of $-\log \Gamma(n + a + b - N + t)$, multiplied by the nonzero constant value. The closed side computes E_N from (4.1)–(4.2), without forming singular polygamma values at $a - N$ or $b - N$.

A.3 Stable evaluation inside the unit disk

A useful exact formula for the regularized generating function is

$$\begin{aligned} F(u, z) = \sum_{n=0}^{L-1} r_n(u) z^n \\ + r_L(u) z^L {}_3F_2 \left(\begin{matrix} 1, L + a, L + b \\ L + 1, L + a + b + u \end{matrix}; z \right), \end{aligned} \quad (\text{A.4})$$

provided L is chosen so that the displayed tail parameters are nonexceptional locally. It follows by shifting the power-series index; the upper parameter 1 cancels the hypergeometric $k!$. This

formula keeps all finite-head reciprocal-gamma zeros and permits ordinary Taylor differentiation of a regular tail. The numerical suite uses it to check the interior identities separately from the coefficient construction.

A.4 Files and commands

The package contains the complete L^AT_EX source, compiled PDF, section sources, and the following principal artifacts:

File	Purpose
<code>code/gauss_hurwitz.py</code>	Exact coefficient jets and numerical evaluators.
<code>code/verify_exact.py</code>	Polynomial and rational regression tests.
<code>code/verify_numeric.py</code>	Independently constructed numerical comparisons.
<code>results/exact_checks.json</code>	Names and status of all finite exact checks.
<code>results/numerical_checks.json</code>	Precision, configurations, values, and residuals.
<code>results/central_binomial_table.json</code>	Exact counterterms and values through $N = 10$.
<code>integration/README.md</code>	Suggested placement and notation requirements.
<code>integration/pslq-wording.patch</code>	Minimal proposed editorial correction.
<code>notes/SOURCE_AUDIT.md</code>	Source provenance and audit limitations.

From the extracted package directory, run

```
python -m pip install -r requirements.txt
python code/verify_exact.py
python code/verify_numeric.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The tested environment was Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The code is intended for Python 3.10 or later; only the recorded environment was executed for this delivery. The build does not require network access once the Python dependencies and TeX packages are installed. The manifest checksums describe the delivered files; rerunning tests updates timing fields in the result files and therefore changes their checksums.

References

- [1] NIST Digital Library of Mathematical Functions, *Chapter 15: Hypergeometric Function*, §15.4, especially Eq. 15.4.20. <https://dlmf.nist.gov/15.4>. Accessed 10 October 2026.
- [2] NIST Digital Library of Mathematical Functions, *Chapter 5: Gamma Function*, §5.11, Asymptotic Expansions. <https://dlmf.nist.gov/5.11>. Accessed 10 October 2026.
- [3] NIST Digital Library of Mathematical Functions, *Chapter 25: Zeta and Related Functions*, §25.11, Hurwitz Zeta Function. <https://dlmf.nist.gov/25.11>. Accessed 10 October 2026.
- [4] NIST Digital Library of Mathematical Functions, *Chapter 25: Zeta and Related Functions*, §25.14, Lerch’s Transcendent. <https://dlmf.nist.gov/25.14>. Accessed 10 October 2026.
- [5] W. Bühring, *Partial sums of hypergeometric series of unit argument*, Proceedings of the American Mathematical Society **132** (2004), 407–415. Preprint arXiv:math/0311126 (2003). <https://arxiv.org/abs/math/0311126>.

- [6] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: part 2*, preprint arXiv:math/0107082 (2001). <https://arxiv.org/abs/math/0107082>.
- [7] H. R. P. Ferguson, D. H. Bailey, and S. Arno, *Analysis of PSLQ, an integer relation finding algorithm*, Mathematics of Computation **68** (1999), no. 225, 351–369. DOI: [10.1090/S0025-5718-99-00995-3](https://doi.org/10.1090/S0025-5718-99-00995-3).
- [8] V. Reshetnikov and ProveIt contributors, *Polylogarithms canonical manuscript README*. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/manuscript>. Inspected 10 October 2026; README blob 02774c39a68fff3792105aed3e2df11e3e604484.
- [9] V. Reshetnikov and ProveIt contributors, *Integration chapter*, canonical manuscript chapter 07-integration.tex. Inspected at commit e0d9463bdee9685dfb1dddb819059cc738540c57; chapter blob 52c8487a618429fb8a0cd3978705614b9a193505. <https://github.com/VladimirReshetnikov/ProveIt>.